

Introduction

Kevin Linares

Joint Program in Survey Methodology, University of Maryland

July 20, 2026

Abstract

Latent class analysis (LCA) of categorical items collected under a stratified, clustered, weighted sample design requires three things that no single R package currently provides together: a design-weighted objective so that estimates describe the population rather than the achieved sample, model selection statistics placed on a scale consistent with that objective, and variance estimation that respects the design in the absence of a usable information matrix. We present an integrated pipeline meeting all three requirements. Parameters are estimated by pseudo-maximum likelihood through a weighted EM algorithm; information criteria are calibrated by weight normalization; and all uncertainty statements derive from stratified delete-one jackknife (JKN) replication, with replicate refits warm-started and aligned to the full-sample solution. Segment composition across external variables is estimated by the BCH three-step correction with design-based replicate intervals, and the unweighted special case is validated to exact agreement against an independent maximum likelihood implementation. We illustrate the methodology with thirteen institutional trust items from the 2023 AmericasBarometer survey of Mexico, identifying five trust segments whose composition varies sharply with presidential approval and satisfaction with democracy. Class naming is drafted by a large language model operating under a certified, frozen prompt instrument with full analyst override.

Keywords: latent class analysis, complex survey data, pseudo-maximum likelihood, replicate variance estimation, three-step estimation, institutional trust

Latent class analysis identifies unobserved population subgroups from response patterns over categorical indicators (Lazarsfeld & Henry 1968, Goodman 1974). When the indicators come from a complex probability sample, three methodological requirements arise jointly. First, the likelihood must be weighted so that the recovered classes and their prevalences describe the target population rather than the achieved sample (Patterson et al. 2002, Asparouhov 2005). Second, model selection statistics must sit on a scale consistent with the weighted objective, since base weights rarely sum to the sample size and the pseudo-log-likelihood is otherwise miscalibrated against the penalty term (Lumley & Scott 2015). Third, variance estimation must respect the stratification and clustering of the design; under a pseudo-likelihood the inverse information matrix is not a valid variance estimator, so design-based replication is the natural device (Wolter 2007, Rust & Rao 1996).

No R package meets these three requirements together with a path from measurement model to labelled analytic data product. The `poLCA` package (Linzer & Lewis 2011) fits unweighted maximum likelihood only. The `survey` package (Lumley 2004) supplies design declaration and replication machinery but no mixture models. The closest existing software, the `baysc` package (Wu et al. 2024, 2026), fits the same weighted objective within a Bayesian pseudo-posterior with a sandwich-based variance adjustment; the pipeline described here is its frequentist counterpart, carrying replicate-based variance through model selection, measurement-model inference, bias-adjusted profiling on external variables, prediction for item-partial respondents, and a labelled data file for downstream analysts. Commercial options exist, notably Mplus with `TYPE = MIXTURE COMPLEX` and Latent GOLD, and Stata’s `gsem` under `svy`; the contribution here is an open, auditable implementation in R in which every estimator corresponds line for line to documented source code, with the unweighted special case validated against an independent implementation.

Throughout, we refer to latent classes as segments in all user-facing output, a terminological preference of applied audiences; the statistical object is unchanged.

We illustrate the methodology with an analysis of institutional trust in Mexico using the 2023 AmericasBarometer (LAPOP Lab 2023). Institutional trust batteries are a natural LCA application: respondents rarely trust or distrust all institutions uniformly, and the substantive interest lies in the configuration of trust across institutions, which a unidimensional scale conceals. Five segments emerge, ranging from generalized distrust to broad confidence, with two intermediate profiles distinguished chiefly by their treatment of the president and the armed forces relative to representative and justice institutions.

Data

The 2023 AmericasBarometer round in Mexico interviewed 1,622 voting-age adults under a stratified, clustered design with 4 strata and 130 primary sampling units, released with base weights (LAPOP Lab 2023). We analyze the thirteen-item institutional trust battery: trust in the justice system, the electoral tribunal, the armed forces, the legislature, the public ministry, the national police, the federal auditor, political parties, the president, the supreme court, the respondent’s municipality, the media, and elections. Each item is answered on a seven-point scale anchored at 1, none at all (*nada*), and 7, a lot (*mucho*); intermediate points are unlabelled. Nonresponse codes are recoded to missing during preparation, and each item is recoded to consecutive integers over its substantive categories.

Estimation uses complete cases on the thirteen items and the profiling covariates: 1,430 of 1,622 respondents (88.2 percent) enter the analysis sample. This is a deliberate design choice rather than an oversight; Section 7 describes how the fitted model then assigns

segment membership to item-partial respondents in the preserved full file, subject to a minimum-information floor, so the complete-case restriction affects estimation but not the coverage of the delivered data product. Profiling covariates are age category, sex, education, urban or rural residence, employment status, satisfaction with democracy, and evaluation of the president’s job performance.

Methods

1 Model

Let $\mathbf{y}_i = (y_{i1}, \dots, y_{iJ})$ denote respondent i ’s responses to J nominal items, item j having C_j categories. With K latent segments, prevalences π_k , and conditional response probabilities $\rho_{jck} = P(y_{ij} = c \mid k)$, the marginal probability of a response pattern is

$$P(\mathbf{y}_i) = \sum_{k=1}^K \pi_k \prod_{j=1}^J \rho_{j y_{ij} k}, \quad \sum_{c=1}^{C_j} \rho_{jck} = 1. \quad (1)$$

Conditional independence of the items given segment membership, local independence, is the load-bearing assumption; Section 5 probes where it strains and Section 7 exploits it for prediction under item nonresponse.

2 Design-weighted pseudo-maximum likelihood

Under a complex design with base weights w_i , estimation maximizes the pseudo-log-likelihood

$$\ell_w(\theta) = \sum_{i=1}^n w_i \log \left(\sum_{k=1}^K \pi_k \prod_{j=1}^J \rho_{j y_{ij} k} \right), \quad (2)$$

the standard device for LCA under complex sampling ([Patterson et al. 2002](#), [Skinner 1989](#)). The EM algorithm ([Dempster et al. 1977](#)) alternates an E-step computing posterior membership probabilities with the log-sum-exp device for numerical stability,

$$p_{ik} = \frac{\pi_k \prod_j \rho_{j y_{ij} k}}{\sum_l \pi_l \prod_j \rho_{j y_{ij} l}}, \quad (3)$$

and an M-step in which the weights enter every sufficient statistic,

$$\hat{\pi}_k = \frac{\sum_i w_i p_{ik}}{\sum_i w_i}, \quad \hat{\rho}_{jck} = \frac{\sum_i w_i p_{ik} \mathbf{1}(y_{ij} = c)}{\sum_i w_i p_{ik}}. \quad (4)$$

Iteration stops when $|\Delta \ell_w| < \varepsilon (|\ell_w| + 1)$. During enumeration $\varepsilon = 10^{-8}$ with at most 800 iterations per start; the chosen model is refit at terminal precision, $\varepsilon = 10^{-10}$ with at most 5,000 iterations, matching the stopping behavior of the reference implementation used for validation so that the comparison in Section 6 contrasts optima rather than stopping rules. Because the pseudo-likelihood of a finite mixture can be multimodal with near-tied optima, each candidate model is fit from twenty random starts with deterministic seeding, and the reported solution is the best mode found; this claim discipline is stated rather than hidden. Point estimates are invariant to the scale of the weights.

3 Model selection

Information criteria compare the maximized log-likelihood against a sample-size penalty, and the two must sit on the same scale. Base weights rarely sum to n ; because ℓ_w is linear in the weights, multiplying it by $n / \sum_i w_i$ is exactly equivalent to fitting with weights rescaled to sum to n , leaving point estimates untouched and calibrating only model selection [[Lumley & Scott \(2015\)](#) develop the general design-based treatment; the sum-to- n

convention matches the default weight normalization in Mplus, making criteria comparable across software]. With $\ell_w^* = \ell_w \cdot n / \sum_i w_i$ and model dimension

$$d_K = (K - 1) + K \sum_{j=1}^J (C_j - 1), \quad (5)$$

we report $\text{BIC} = -2\ell_w^* + d_K \log n$ and $\text{AIC} = -2\ell_w^* + 2d_K$, alongside the relative entropy

$$R_E^2 = 1 - \frac{-\sum_{i,k} p_{ik} \log p_{ik}}{n \log K}, \quad (6)$$

which indexes classification sharpness (Ramaswamy et al. 1993). The bootstrapped likelihood ratio test, often recommended for class enumeration under simple random sampling (Nylund et al. 2007), is deliberately omitted: its parametric bootstrap presumes independent observations, which fails under a design-weighted pseudo-likelihood on a clustered sample, matching the equivalent restriction imposed by Mplus under complex-design analysis. The number of segments is never selected automatically. The analyst chooses K by iterating over these statistics and the diagnostics of Section 5, and fixes it in configuration; Section 10 reports that reasoning for the present application.

4 Replicate variance estimation

A pseudo-likelihood provides no usable information matrix, so all uncertainty statements come from stratified delete-one-PSU jackknife (JKn) replication (Wolter 2007, Rust & Rao 1996, Lumley 2004). With n_h PSUs in stratum h , deleting PSU j of stratum h and reweighting its stratum-mates by $n_h/(n_h - 1)$ yields the replicate estimate $\hat{\theta}_{(hj)}$, and

$$\widehat{V}(\hat{\theta}) = \sum_h \frac{n_h - 1}{n_h} \sum_{j=1}^{n_h} (\hat{\theta}_{(h,j)} - \hat{\theta})^2. \quad (7)$$

Each replicate refits the weighted EM on the replicate weights, warm-started from the full-sample solution, so the resulting variance is within-mode by construction: it carries design variance, not uncertainty about mode selection. Label switching across replicates is resolved by Hungarian assignment (Kuhn 1955, Hornik 2005) on the conditional response profiles, aligning every replicate to the full-sample fit before differencing.

Confidence intervals for every probability, the prevalences π_k and the conditional probabilities ρ_{jck} , are formed on the logit scale with the delta-method standard error $\text{se}_{\text{logit}} = \text{se}/\{\hat{p}(1 - \hat{p})\}$ and back-transformed, which keeps them inside $(0, 1)$ without truncation and improves finite-sample coverage away from one half. The critical value is $t_{0.975, \nu}$ with ν the design degrees of freedom, the number of PSUs minus the number of strata, here $130 - 4 = 126$, rather than the normal 1.96. Cells estimated within 10^{-3} of the boundary have no meaningful logit-scale standard error and instead receive Wald intervals truncated to $[0, 1]$, flagged as such in the delivered tables. The same interval construction serves the domain estimates of Section 8, so a single method runs through the entire pipeline.

5 Model-fit diagnostics

Item discrimination summarizes how much item j separates the segments as the mean, over segment pairs, of the total variation distance between their conditional response distributions:

$$D_j = \frac{2}{K(K-1)} \sum_{k < l} \frac{1}{2} \sum_{c=1}^{C_j} |\rho_{jck} - \rho_{jcl}|. \quad (8)$$

$D_j = 0$ means every segment answers item j identically, a removal candidate for a sensitivity refit; values near 1 indicate sharp separation.

Discrimination is an effect size; a companion screen asks whether each item earns its class-dependent parameters by BIC. For item j , the fitted model is compared on the same respondents against a restricted model in which item j is made class-independent, its response distribution shared across segments. Under local independence the restricted model factorizes into an LCA on the remaining items times item j 's weighted marginal, so its pseudo-log-likelihood is available without a constrained refit as $\ell_{-j} + \ell_j^{\text{marg}}$, its dimension is $d_K^{(-j)} + (C_j - 1)$, and its BIC on the sum-to- n scale is directly comparable to the full model's. We report $\Delta\text{BIC}_j = \text{BIC}_{\text{restricted}} - \text{BIC}_{\text{full}}$; a negative value means BIC prefers dropping item j 's class dependence and flags the item.

Local independence is probed pairwise with a bivariate residual (BVR). For items a, b , let $\widehat{P}_{ab}(c, d)$ be the design-weighted observed proportion of the joint response (c, d) and $\Pi_{ab} = \rho_a \text{diag}(\pi) \rho_b^\top$ the model-implied joint distribution. The statistic

$$\text{BVR}_{ab} = \frac{1}{(C_a - 1)(C_b - 1)} n \sum_{c,d} \frac{(\widehat{P}_{ab}(c, d) - \Pi_{ab}(c, d))^2}{\Pi_{ab}(c, d)} \quad (9)$$

is a Pearson-type residual per degree of freedom. In unweighted maximum likelihood settings its null distribution is already not chi-square and bootstrap calibration is advised (Oberski et al. 2013); under weighting and pseudo-likelihood no reference distribution is valid, so we report BVR strictly as a descriptive ranking of the item pairs the model explains least well, with large persistent values arguing for item removal, merging, or $K + 1$.

6 Validation against an independent implementation

At unit weights, Equation 2 reduces to the ordinary log-likelihood, so the weighted EM and `poLCA` (Linzer & Lewis 2011) maximize the same function and must agree up to optimizer tolerance and mode selection. The pipeline enforces this as a runtime check. The unit-weight EM solution is obtained from the same multi-start protocol plus one additional candidate warm-started at the `poLCA` solution itself; since both maximize the identical function, whichever basin attains the higher log-likelihood is retained. The retained solution is aligned to the `poLCA` solution by Hungarian matching on response profiles, and maximum absolute gaps in prevalences and conditional probabilities are computed and frozen to disk with a single pass or fail verdict at a 0.005 threshold on both gaps, printed identically in every downstream document. Where a gap appears, comparison of the two log-likelihoods separates flat-ridge drift under the stopping rule, an inferior `poLCA` mode, and an inferior EM mode calling for more starts. The same reasoning positions the pipeline against Mplus (`TYPE = MIXTURE COMPLEX`) and Stata (`gsem` under `svy`): identical objective and weight convention imply matching point estimates and criteria up to mode selection, with linearization rather than replication standard errors, both design-consistent.

7 Segment assignment under partial response

Membership is assigned from the posterior evaluated at the fitted parameters. Local independence makes assignment honest for item-partial respondents: a missing item drops out of the within-segment product, so for respondent i with answered item set A_i ,

$$p_{ik}^{\text{obs}} = \frac{\pi_k \prod_{j \in A_i} \rho_{j y_{ij} k}}{\sum_l \pi_l \prod_{j \in A_i} \rho_{j y_{ij} l}}, \quad (10)$$

and the assigned segment is $\arg \max_k p_{ik}^{\text{obs}}$. An information floor m^* , configured at seven of the thirteen items here, is enforced: respondents answering fewer than m^* items receive no assignment rather than a weakly informed one. The model is fit on complete cases only; assignment extends to the preserved full frame, and the exported labelled data record which respondents were assigned.

8 Segment composition on external variables: the BCH correction

Cross-classifying modal assignments against external variables understates segment differences because assignment error, even at high entropy, attenuates composition toward the marginal (Bolck et al. 2004). We therefore estimate segment prevalence within demographic domains by the BCH three-step correction (Bolck et al. 2004, Vermunt 2010), adapted to the design-weighted setting. Step one is the fitted measurement model. Step two computes the modal assignment $W_i = \arg \max_k p_{ik}$ and the design-weighted classification-error matrix $\mathbf{D}$ with entries

$$D_{ks} = \widehat{P}(W = s \mid X = k) = \frac{\sum_i w_i p_{ik} \mathbf{1}(W_i = s)}{\sum_i w_i p_{ik}}, \quad (11)$$

the weighted posterior mass of true segment k landing in assigned segment s ; rows of $\mathbf{D}$ sum to one. Step three replaces each respondent's hard assignment with the correction vector $\mathbf{u}_i = \mathbf{e}_{W_i}^\top \mathbf{D}^{-1}$, row W_i of $\mathbf{D}^{-1}$. Weighted by w_i , the $\mathbf{u}_i$ are unbiased for true-segment membership, so the BCH-corrected prevalence of segment k within domain d is the design-based domain mean

$$\hat{p}_{dk} = \frac{\sum_{i \in d} w_i u_{ik}}{\sum_{i \in d} w_i}. \quad (12)$$

Two properties of the correction deserve note. Entries of $\mathbf{u}_i$ can be negative, so individual cell estimates can fall slightly outside $[0, 1]$; this is a documented property of the method, not an error. And each $\mathbf{u}_i$ sums to one because the rows of $\mathbf{D}$ do, so within every domain the estimated prevalences still sum to one across segments.

Variance is estimated with the same JK_n replicates: the posterior matrix and modal assignments are held fixed at the full-sample measurement model, while $\mathbf{D}$ and the correction weights are rebuilt inside every replicate from the replicate weights, so the intervals carry the design variance of the classification table and the cross-tabulation jointly. Holding step one fixed is the standard treatment; the resulting understatement of variance is minor when classification is sharp (Bakk et al. 2014, 2013), and the present application, with relative entropy 0.907, is squarely in that favorable regime. Intervals follow Section 4, logit-scale with the design t critical value; cells outside $(0, 1)$, where the logit transform is undefined, fall back to truncated Wald intervals and are footnoted in the delivered tables.

9 LLM-assisted, analyst-controlled labeling

Segment naming is a bottleneck in applied LCA reporting, and recent evidence suggests large language models perform annotation-adjacent tasks reliably when tightly instructed (Gilardi et al. 2023, Ziems et al. 2024). The pipeline drafts segment names with a language model under a fixed instrument: a persona and rule set instructing interpretation strictly from the displayed conditional response probabilities and item wording, one isolated call per segment (joint prompts empirically confused near-neighbor segments), an optional survey-

context paragraph fenced to referent resolution only, and a collision-gated harmonization pass that may edit labels, never descriptions. Labels follow one rule: a label file on disk is used when present and validated against K ; otherwise the model drafts once and the result is frozen to that file, which the analyst edits to take over naming. The freeze file, not a sampling seed, is the reproducibility mechanism.

The instrument is treated as a measurement device. A certification harness plants a known-meaning model, including near-neighbor segments, a deliberately diffuse segment, nonresponse-labelled items, and a fictional survey context, and checks output format, grounding in the displayed probabilities, hedging on diffuse profiles, persona integrity, context-leak fencing, harmonizer properties, and adoption of the segment terminology; any edit to the instrument obliges recertification. Labels are drafts, verified by the analyst against the conditional response profiles before use, and the delivered tables carry the label file’s contents, not live model output.

Results

10 Enumeration and the choice of K

Table 1 reports the weighted fit statistics for $K = 2$ through 12, each fit from twenty seeded random starts; all candidates converged. Figure 1 plots weighted BIC and relative entropy.

Table 1: Weighted fit statistics by number of segments. The log-likelihood is on the sum-to- n scale; entropy is the relative entropy of the posterior classification.

K	Log-likelihood	df	AIC	BIC	Entropy
2	-31579.7	157	63473.3	64300.0	0.904

Table 1: Weighted fit statistics by number of segments. The log-likelihood is on the sum-to-n scale; entropy is the relative entropy of the posterior classification.

K	Log-likelihood	df	AIC	BIC	Entropy
3	-30176.2	236	60824.3	62067.0	0.913
4	-29447.7	315	59525.5	61184.1	0.921
5	-29047.7	394	58883.4	60957.9	0.907
6	-28751.8	473	58449.5	60940.1	0.895
7	-28537.7	552	58179.3	61085.8	0.893
8	-28390.2	631	58042.5	61365.0	0.890
9	-28242.7	710	57905.4	61643.8	0.890
10	-28123.2	789	57824.4	61978.8	0.894
11	-28024.0	868	57784.0	62354.4	0.902
12	-27927.8	947	57749.6	62736.0	0.899

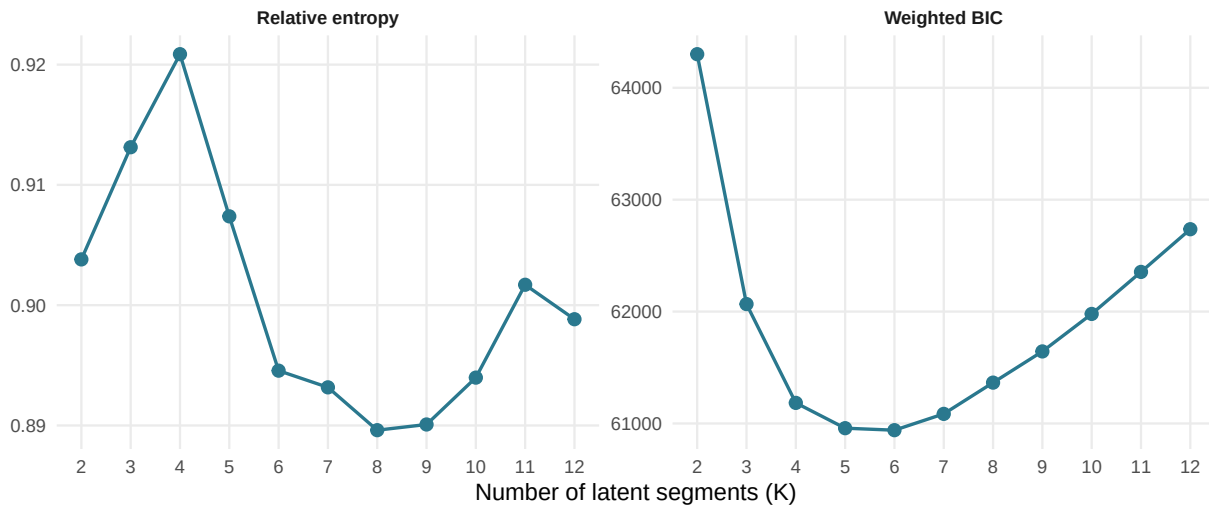


Figure 1: Enumeration diagnostics across K: weighted BIC and relative entropy. BIC is essentially flat between $K = 5$ and $K = 6$; entropy peaks at $K = 4$ and declines thereafter.

Weighted BIC drops steeply through $K = 5$ (60,957.9), reaches its minimum at $K = 6$ (60,940.1), and rises monotonically thereafter. The difference between the two, 17.8, is small against the preceding drops of 226.2 ($4 \rightarrow 5$) and 882.9 ($3 \rightarrow 4$): the criterion is effectively flat over the $\{5, 6\}$ region. AIC continues to decline through the whole range, its usual overextraction under large n (Nylund et al. 2007). Relative entropy peaks at $K = 4$ (0.921) and remains high at $K = 5$ (0.907); no larger K exceeds 0.91. Within the flat BIC region we fixed $K = 5$ on parsimony, classification sharpness, and interpretability: the five-segment solution yields substantively distinct, cleanly separated profiles (Section 11), whereas the marginal BIC gain at $K = 6$ purchases a further split at the cost of entropy. Consistent with the flatness of the criterion, the unweighted benchmark’s own BIC minimizes at $K = 6$ as well; enumeration under weighting and without it point to the same region. We emphasize that K is an analyst decision recorded in configuration, not an automated selection, and the full enumeration evidence is reported so readers can weigh the neighboring solution.

11 The measurement model

The chosen model converged with weighted log-likelihood $-29,047.7$ on the sum-to- n scale, relative entropy 0.907, and mean maximum posterior probability 0.939: respondents are, on average, assigned to their modal segment with 94 percent confidence. Table 2 reports the design-weighted segment prevalences with JK_n intervals alongside the unweighted benchmark, and Figure 2 displays the full measurement model, the estimated ρ_{jck} for every item, category, and segment.

Table 2: Design-weighted segment prevalences with 95 percent JK_n intervals (logit scale, t critical value with 126 design degrees of freedom), the unweighted poLCA benchmark, and the weighting gap.

Segment	Weighted prevalence	Unweighted	
	[95% CI]	benchmark	Gap
S1: Moderate Institutional Trust	0.331 [0.293, 0.372]	0.336	-0.005
S2: Generalized Institutional Distrust	0.107 [0.086, 0.132]	0.107	+0.000
S3: Presidential and Military Trust	0.242 [0.204, 0.285]	0.239	+0.004
S4: Very High National Trust	0.111 [0.093, 0.134]	0.111	-0.000
S5: Broad Institutional Trust	0.208 [0.173, 0.249]	0.207	+0.001

The five profiles order naturally on overall trust with two configurational departures. Segment S5, Broad Institutional Trust (weighted prevalence 0.208), concentrates mass on categories 5 and 6 across every institution. Segment S4, Very High National Trust (0.111), is its extreme counterpart, with probability 0.82 of answering *mucho* on trust in the president and 0.70 or above on the public ministry, armed forces, supreme court, and electoral tribunal. Segment S2, Generalized Institutional Distrust (0.107), mirrors it at the other pole, with probability 0.85 of answering *nada* on political parties and 0.76 on the national police; only the president and armed forces draw any appreciable mass off the floor. The two largest segments are configurational rather than uniform. Segment S1, Moderate Institutional Trust (0.331), holds the middle categories on the justice and audit institutions while shifting upward on the president and armed forces. Segment S3, Presidential and

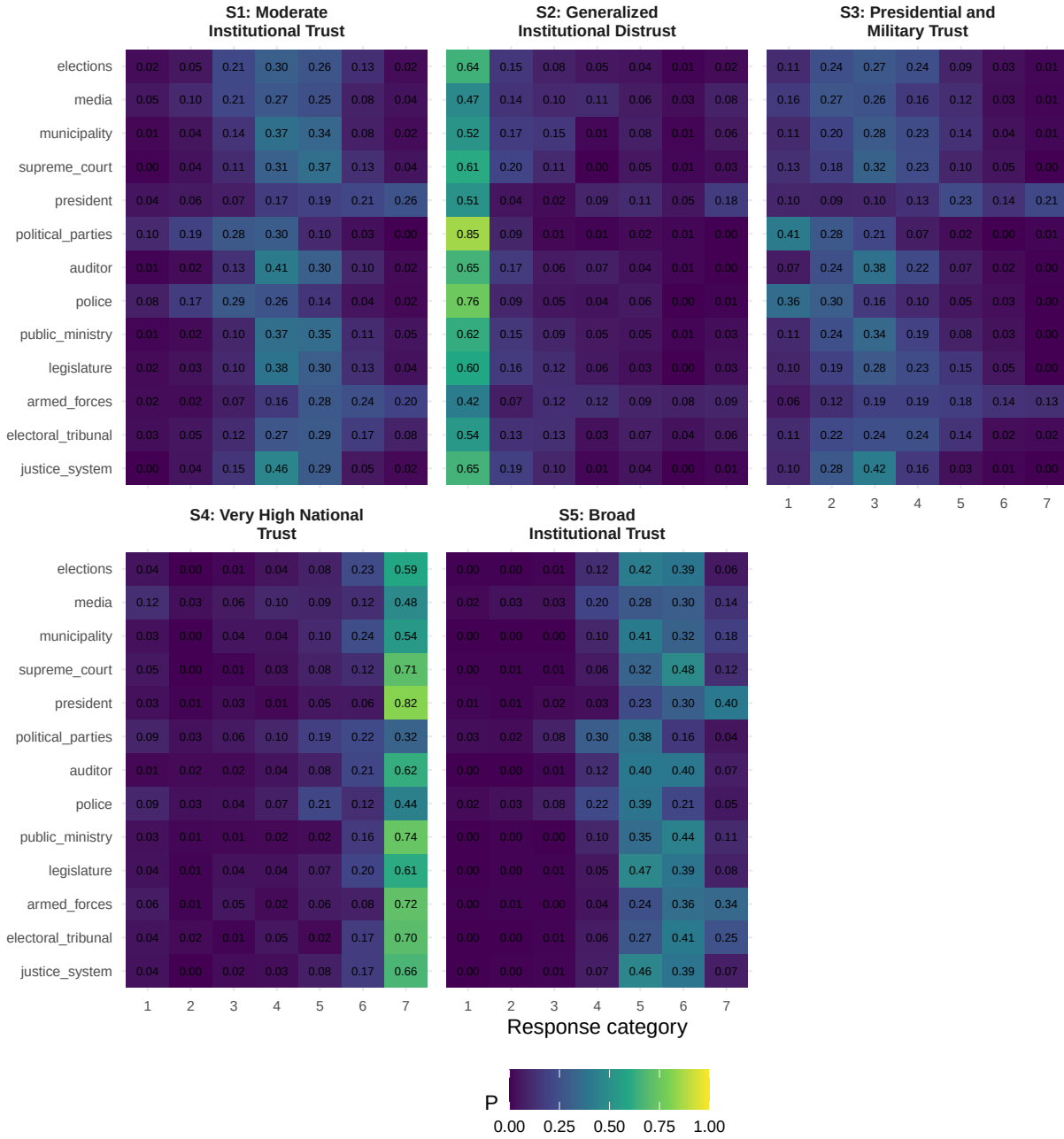


Figure 2: Conditional response probabilities $P(\text{response category} | \text{segment})$ for all thirteen trust items. Rows are items, columns are the seven response categories from 1 (nada) to 7 (mucho).

Military Trust (0.242), sharpens that asymmetry: substantial trust in the president (probability 0.21 at *mucho*, mass concentrated at 5 to 7) and the armed forces, against firm distrust of political parties (0.41 at *nada*) and the police. The configuration, not the level, separates S1 from S3, and it is precisely this configurational structure that a summed trust scale would average away.

The weighted and unweighted prevalences differ by at most 0.005 (Table 2): the released weights for this single-country file are nearly constant, so weighting shifts point estimates little here. The design still matters for inference: the JK_n intervals reflect the clustering, and the interval for S1, for example, spans 0.293 to 0.372, wider than an SRS binomial interval at $n = 1,430$ would suggest.

12 Diagnostics

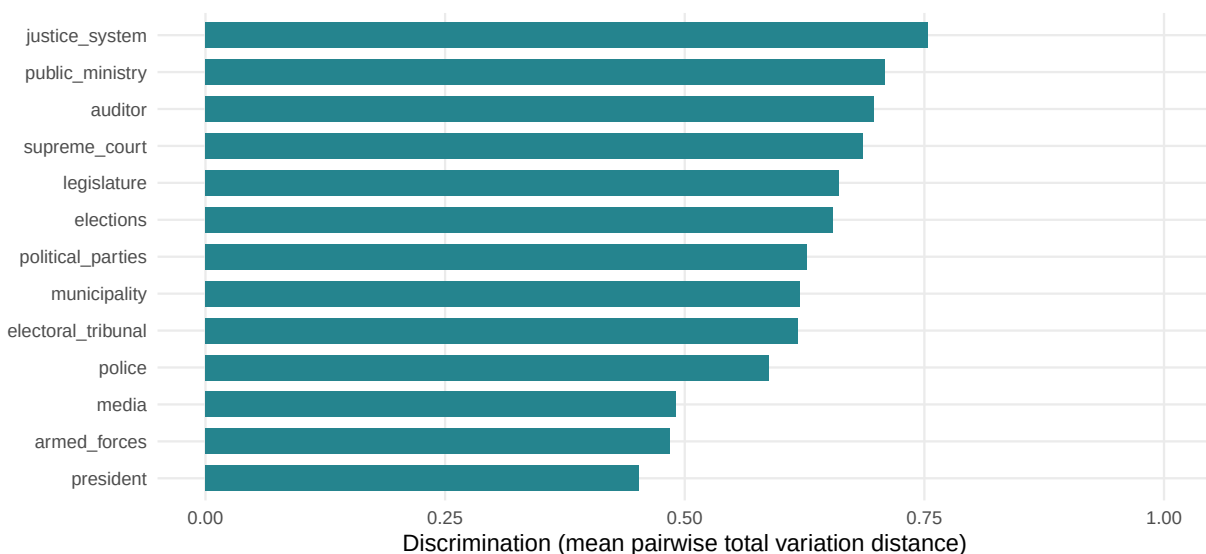


Figure 3: Item discrimination: the mean over segment pairs of the total variation distance between conditional response distributions.

Every item earns its place. Discrimination (Figure 3) ranges from 0.452 (president) to

0.754 (justice system); no item approaches the region where removal would be considered. The BIC screen of Section 5 agrees: ΔBIC_j is positive for all thirteen items, from +320.0 (president) to +1,419.3 (justice system), so BIC prefers retaining every item’s class dependence, by margins that are decisive throughout. That the president and armed forces items discriminate least is itself informative: they are the institutions on which otherwise different segments agree.

Table 3: Largest bivariate residuals. BVR is a descriptive ranking under the weighted pseudo-likelihood; no chi-square reference distribution applies.

Item A	Item B	Pair df	BVR
public_ministry	auditor	36	5.75
armed_forces	legislature	36	5.54
justice_system	supreme_court	36	3.21
police	political_parties	36	2.81
president	supreme_court	36	2.48

The bivariate residuals (Table 3) identify two item pairs standing above the rest: public ministry with auditor (5.75) and armed forces with legislature (5.54), against a table otherwise at or below 3.21. These are institutionally adjacent pairs, two audit-and-prosecution bodies and two national-state bodies, whose residual association the five segments do not fully absorb. Because no reference distribution is valid here, we read these as a ranked caution rather than a rejection, one consistent with BIC’s mild preference for the neighboring six-segment solution, and we flag the two pairs as the natural first check in any sensitivity refit.

13 Validation and assignment coverage

At unit weights the pipeline’s EM and `poLCA` agree exactly: the maximum absolute prevalence gap is 0.00000, the maximum conditional-probability gap is 0.00000, and the two log-likelihoods are identical to reported precision (-29046.7917). The implementations maximize the same function and find the same optimum, which certifies the weighted estimator’s machinery before any weighted result is interpreted.

Prediction under Equation 10 assigned a segment to 1,618 of 1,622 respondents: all 1,430 complete cases plus 188 item-partial respondents answering at least the seven-item floor. Four respondents fell below the floor and were left unassigned rather than guessed. The delivered data product is a labelled SPSS file carrying the numeric segment with value labels, the label text, and the assignment diagnostics, together with CSV exports of the full measurement model with intervals and the demographic composition estimates.

14 Demographic composition

BCH-corrected composition was estimated for seven covariates; we highlight the two with the sharpest structure, presidential job approval and satisfaction with democracy, and summarize the remainder. Full tables for all covariates accompany the replication materials.

Table 4: BCH-corrected segment prevalence within presidential job approval, with 95 per cent JK_n intervals on the logit scale. Rows sum to one across segments. One cell (Bad approval, S4) has a slightly negative BCH point estimate, a documented property of the correction; its interval is truncated Wald.

Level	S1	S2	S3	S4	S5
Very bad	0.21 [0.11, 0.38]	0.52 [0.35, 0.68]	0.14 [0.05, 0.33]	0.05 [0.01, 0.33]	0.08 [0.02, 0.24]
Bad	0.30 [0.19, 0.44]	0.31 [0.20, 0.44]	0.33 [0.21, 0.49]	-0.00 [0.00, -0.00]	0.06 [0.02, 0.17]
Neither good nor bad	0.39 [0.32, 0.46]	0.16 [0.13, 0.20]	0.26 [0.21, 0.32]	0.07 [0.04, 0.10]	0.12 [0.08, 0.18]
Good	0.33 [0.30, 0.37]	0.06 [0.04, 0.09]	0.24 [0.21, 0.28]	0.10 [0.08, 0.13]	0.26 [0.22, 0.30]
Very good	0.30 [0.25, 0.35]	0.07 [0.05, 0.10]	0.21 [0.17, 0.26]	0.19 [0.16, 0.24]	0.23 [0.18, 0.28]

Composition is strongly ordered in presidential approval (Table 4, Figure 4). Among respondents rating the president’s performance very bad, Generalized Institutional Distrust reaches 0.52 [0.35, 0.68]; among those rating it very good, that segment falls to 0.07 while Very High National Trust rises to 0.19 [0.16, 0.24]. The correction’s behavior at sparse cells is visible in the Bad approval row, where the S4 estimate is a numerically negative zero with a truncated interval: no respondents rating the president badly belong, after correction, to the segment defined by near-uniform *mucho* responses that include the president

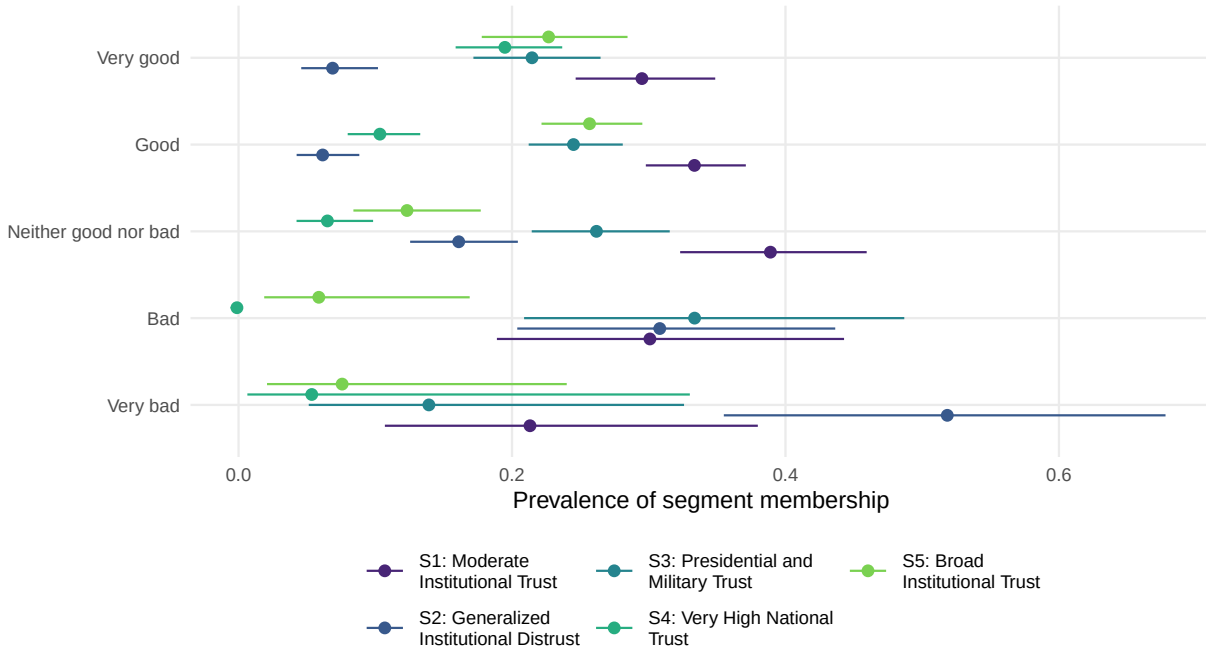


Figure 4: BCH-corrected segment composition by presidential job approval (95 percent JK n intervals).

item, exactly as coherence requires. Satisfaction with democracy behaves symmetrically: Generalized Distrust is 0.34 [0.25, 0.45] among the very dissatisfied against 0.06 among the very satisfied, while Very High National Trust is 0.34 [0.25, 0.44] among the very satisfied against 0.05 among the very dissatisfied.

The remaining covariates show more modest structure. Presidential and Military Trust is more prevalent in urban areas (0.26 [0.23, 0.29]) than rural (0.18 [0.14, 0.22]) and declines with age, from 0.25 among respondents 18 to 29 to 0.17 among those 60 and over, while Broad Institutional Trust moves in the opposite direction with age. Respondents with no formal education are estimated predominantly in Very High National Trust (0.41 [0.22, 0.64]), though this cell rests on few observations and its interval is correspondingly wide. Differences by sex and employment status are small, with overlapping intervals throughout.

Discussion

The pipeline demonstrated here closes a specific gap: design-weighted latent class estimation with fully integrated, design-based replicate inference, in open R code, validated against an independent implementation, and carried through to bias-adjusted profiling and a labelled data product. Each component is individually established, pseudo-maximum likelihood for LCA under complex sampling ([Patterson et al. 2002](#)), weight-normalized information criteria ([Lumley & Scott 2015](#)), JK_n replication ([Wolter 2007](#)), and the BCH correction ([Bolck et al. 2004](#), [Vermunt 2010](#)); the contribution is their disciplined integration, with the seams between components, alignment of replicate fits, a single frozen validation verdict, one interval construction throughout, treated as first-class design decisions rather than afterthoughts.

The application illustrates why the machinery matters substantively. Institutional trust in Mexico is not unidimensional: two of the five segments, together a third of the population, are defined by the configuration of trust, elevated for the president and armed forces against distrust of parties and police, rather than by its level, and segment composition moves sharply with presidential approval. A summed trust scale would compress exactly this structure. The near-constancy of the released weights in this file makes the weighting correction small for point estimates while leaving the design correction to inference fully operative; in surveys with more variable weights the same pipeline applies unchanged with larger weighting gaps.

Several limitations bound the claims. The pseudo-likelihood can be multimodal with near-tied optima; the reported solution is the best mode under many seeded starts, and JK_n replication expresses within-mode design variance only. The BCH step conditions on the fitted measurement model; with relative entropy 0.907 the resulting understatement of variance is minor ([Bakk et al. 2014](#)), but it is an understatement. Estimation uses complete cases,

with model-based assignment rather than imputation extending results to item-partial respondents above the information floor. The choice of K is an argued analyst decision within a flat region of the criterion, and we report the full enumeration evidence precisely so that readers can weigh the neighboring six-segment solution. Finally, language-model-drafted labels are a documented convenience with a certified instrument and full analyst override, not a measurement claim; all substantive interpretation rests on the estimated conditional response probabilities.

Replication materials, including all source code, the rendered analysis documents, and the frozen output objects from which every table and figure in this article is generated, are available in the project repository.

References

- Asparouhov, T. (2005), ‘Sampling weights in latent variable modeling’, *Structural Equation Modeling* **12**(3), 411–434.
- Bakk, Z., Oberski, D. L. & Vermunt, J. K. (2014), ‘Relating latent class assignments to external variables: Standard errors for correct inference’, *Political Analysis* **22**(4), 520–540.
- Bakk, Z., Tekle, F. B. & Vermunt, J. K. (2013), ‘Estimating the association between latent class membership and external variables using bias-adjusted three-step approaches’, *Sociological Methodology* **43**(1), 272–311.
- Bolck, A., Croon, M. & Hagnaars, J. (2004), ‘Estimating latent structure models with categorical variables: One-step versus three-step estimators’, *Political Analysis* **12**(1), 3–27.

- Dempster, A. P., Laird, N. M. & Rubin, D. B. (1977), ‘Maximum likelihood from incomplete data via the EM algorithm’, *Journal of the Royal Statistical Society, Series B* **39**(1), 1–38.
- Gilardi, F., Alizadeh, M. & Kubli, M. (2023), ‘ChatGPT outperforms crowd workers for text-annotation tasks’, *Proceedings of the National Academy of Sciences* **120**(30), e2305016120.
- Goodman, L. A. (1974), ‘Exploratory latent structure analysis using both identifiable and unidentifiable models’, *Biometrika* **61**(2), 215–231.
- Hornik, K. (2005), ‘A CLUE for CLUster Ensembles’, *Journal of Statistical Software* **14**(12), 1–25.
- Kuhn, H. W. (1955), ‘The hungarian method for the assignment problem’, *Naval Research Logistics Quarterly* **2**(1–2), 83–97.
- LAPOP Lab (2023), ‘Americasbarometer 2023: Mexico. technical information and country dataset’, <https://www.vanderbilt.edu/lapop/>.
- Lazarsfeld, P. F. & Henry, N. W. (1968), *Latent Structure Analysis*, Houghton Mifflin, Boston.
- Linzer, D. A. & Lewis, J. B. (2011), ‘poLCA: An R package for polytomous variable latent class analysis’, *Journal of Statistical Software* **42**(10), 1–29.
- Lumley, T. (2004), ‘Analysis of complex survey samples’, *Journal of Statistical Software* **9**(8), 1–19.
- Lumley, T. & Scott, A. (2015), ‘AIC and BIC for modeling with complex survey data’, *Journal of Survey Statistics and Methodology* **3**(1), 1–18.
- Nylund, K. L., Asparouhov, T. & Muthén, B. O. (2007), ‘Deciding on the number of classes

- in latent class analysis and growth mixture modeling: A Monte Carlo simulation study', *Structural Equation Modeling* **14**(4), 535–569.
- Oberski, D. L., van Kollenburg, G. H. & Vermunt, J. K. (2013), 'A Monte Carlo evaluation of three methods to detect local dependence in binary data latent class models', *Advances in Data Analysis and Classification* **7**(3), 267–279.
- Patterson, B. H., Dayton, C. M. & Graubard, B. I. (2002), 'Latent class analysis of complex sample survey data: Application to dietary data', *Journal of the American Statistical Association* **97**(459), 721–728.
- Ramaswamy, V., DeSarbo, W. S., Reibstein, D. J. & Robinson, W. T. (1993), 'An empirical pooling approach for estimating marketing mix elasticities with PIMS data', *Marketing Science* **12**(1), 103–124.
- Rust, K. F. & Rao, J. N. K. (1996), 'Variance estimation for complex surveys using replication techniques', *Statistical Methods in Medical Research* **5**(3), 283–310.
- Skinner, C. J. (1989), Domain means, regression and multivariate analysis, in C. J. Skinner, D. Holt & T. M. F. Smith, eds, 'Analysis of Complex Surveys', Wiley, Chichester, pp. 59–87.
- Vermunt, J. K. (2010), 'Latent class modeling with covariates: Two improved three-step approaches', *Political Analysis* **18**(4), 450–469.
- Wolter, K. M. (2007), *Introduction to Variance Estimation*, 2nd edn, Springer, New York.
- Wu, S. M., Williams, M. R., Savitsky, T. D. & Stephenson, B. J. K. (2024), 'Derivation of outcome-dependent dietary patterns for low-income women obtained from survey data using a supervised weighted overfitted latent class analysis', *Biometrics* **80**(4), ujae122.

- Wu, S. M., Williams, M. R., Savitsky, T. D. & Stephenson, B. J. K. (2026), ‘baysc: An R package for Bayesian survey clustering’, *Journal of Open Source Software* **11**(119), 8382.
- Ziems, C., Held, W., Shaikh, O., Chen, J., Zhang, Z. & Yang, D. (2024), ‘Can large language models transform computational social science?’, *Computational Linguistics* **50**(1), 237–291.